

A thick black L-shaped frame surrounds the text. It starts at the top left, goes right, then down, then right again at the bottom.

EXPLORING SUPERCONDUCTING PROPERTIES IN HBN- PROTECTED BSCCO DEVICES

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OUTLINE

MOTIVATION

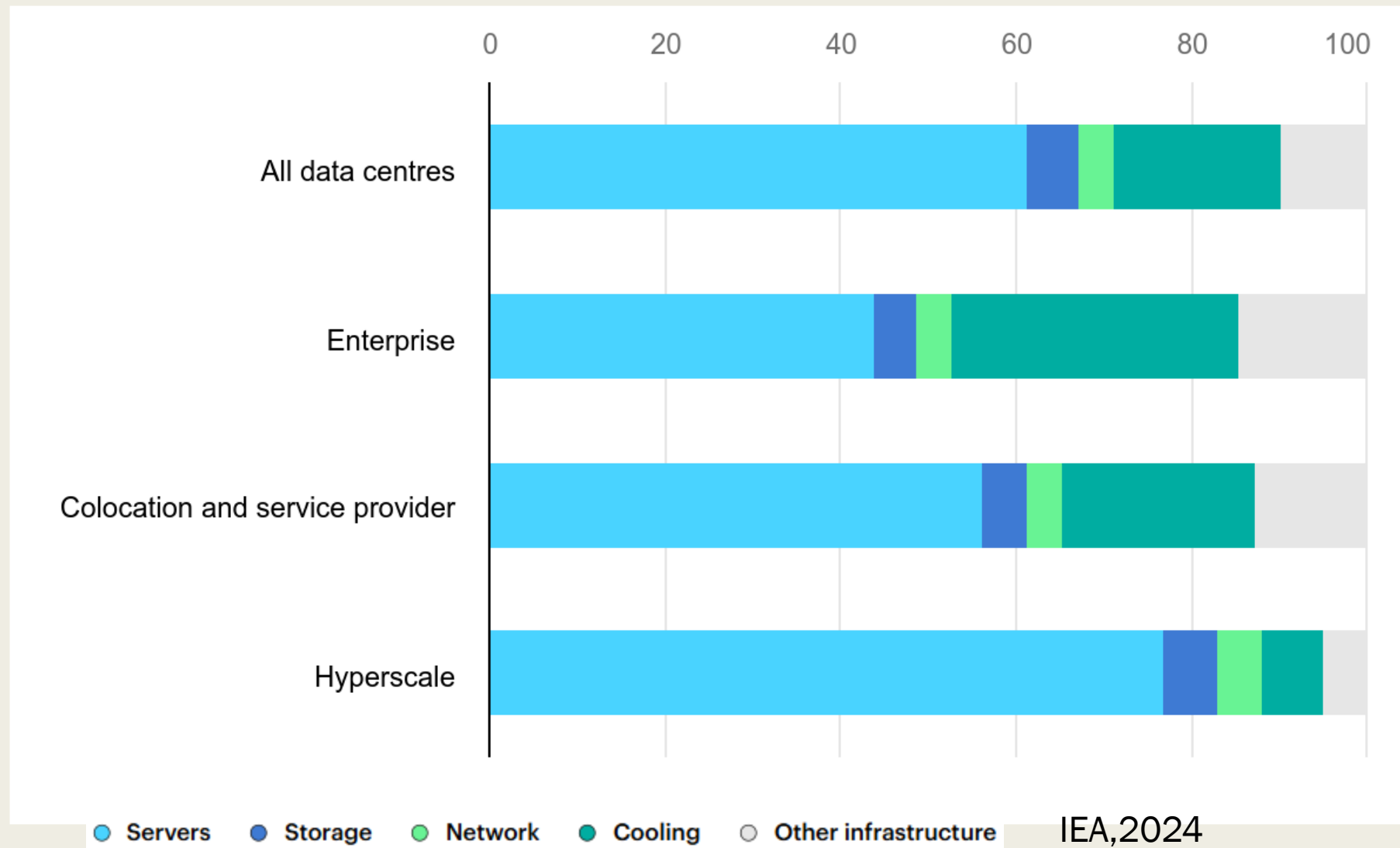
INTRODUCTION

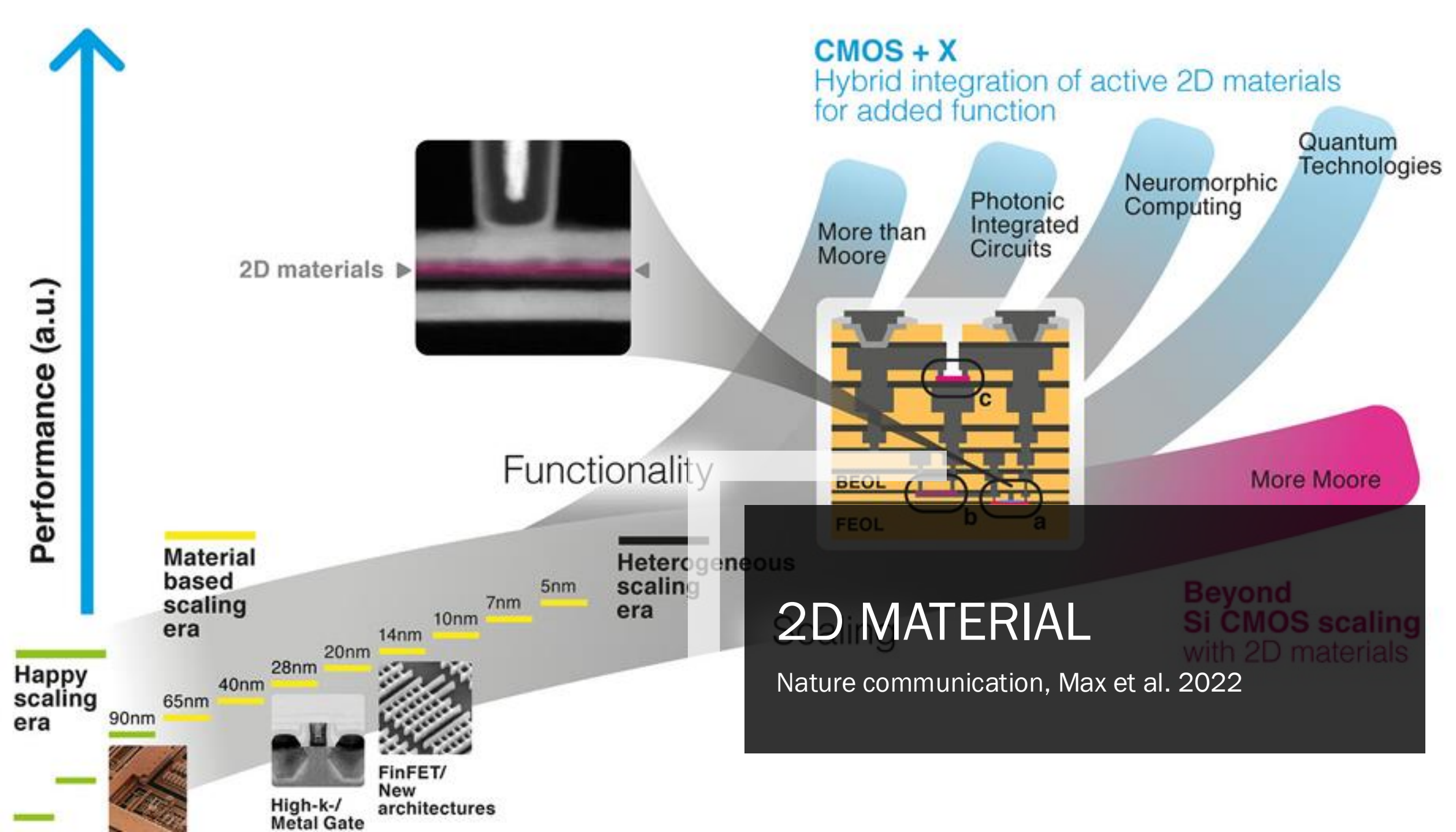
DEVICE FABRICATION

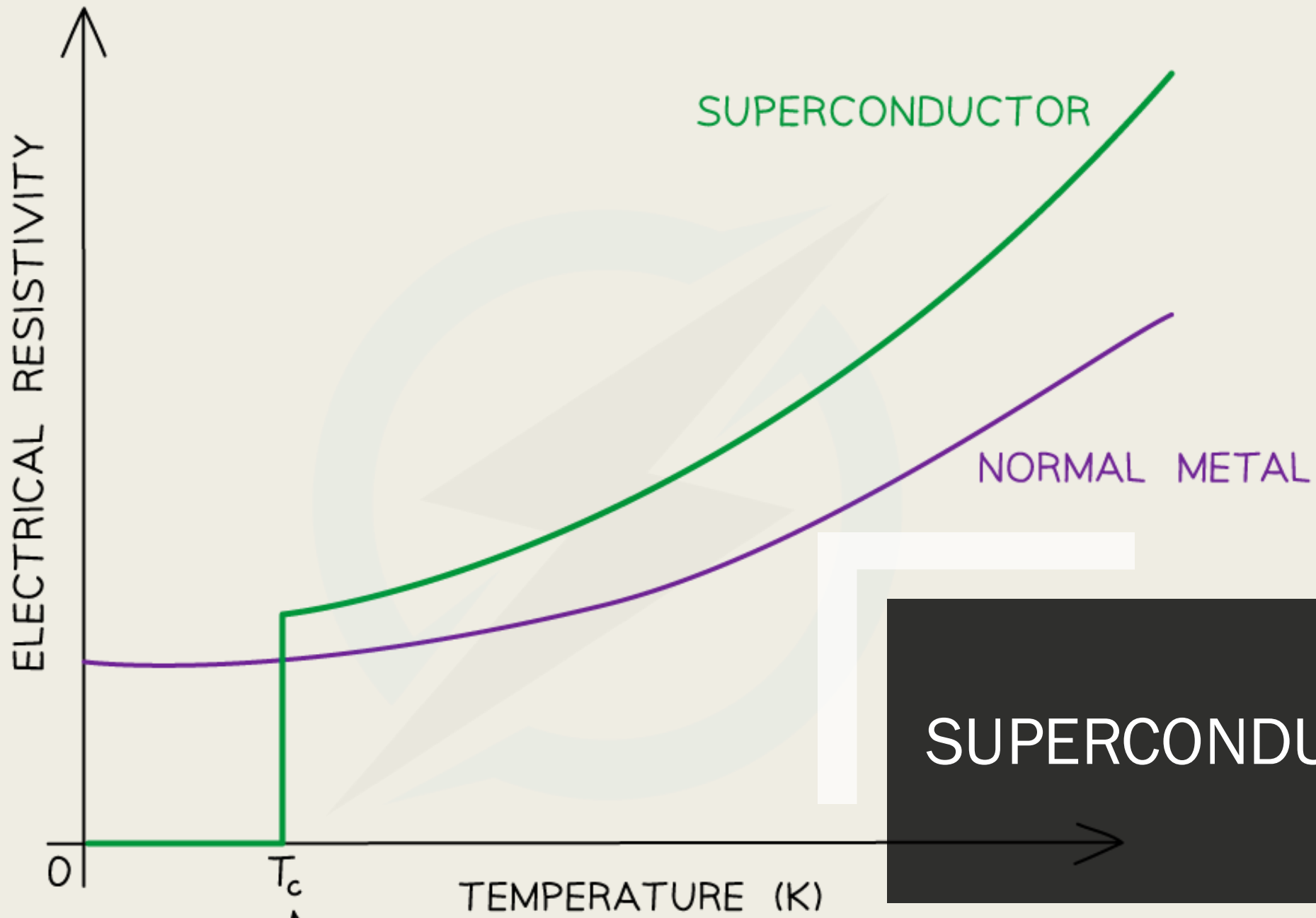
MEASUREMENTS AND ANALYSIS

SUMMARY

$$P = I^2 R$$



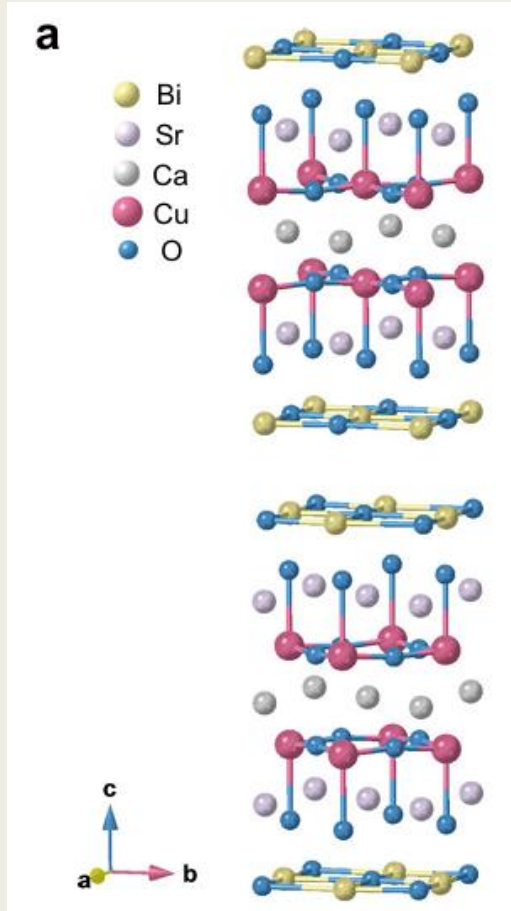




SUPERCONDUCTING

CRITICAL TEMPERATURE

$\text{BiSr}_2\text{CaCu}_2\text{O}_{8+\delta}$ (BSCCO – 2212)

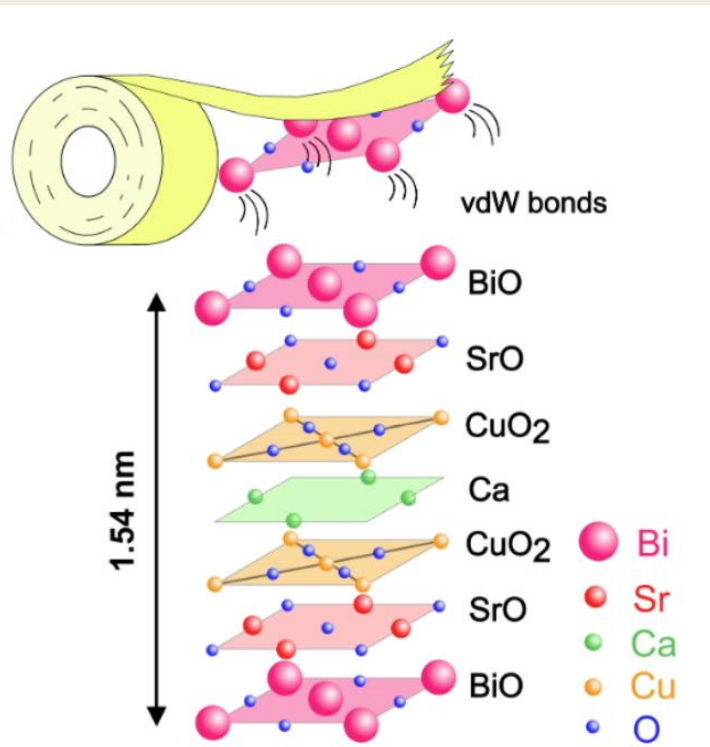


Nature comm, Shichao et al.(2025)

BSCCO problem

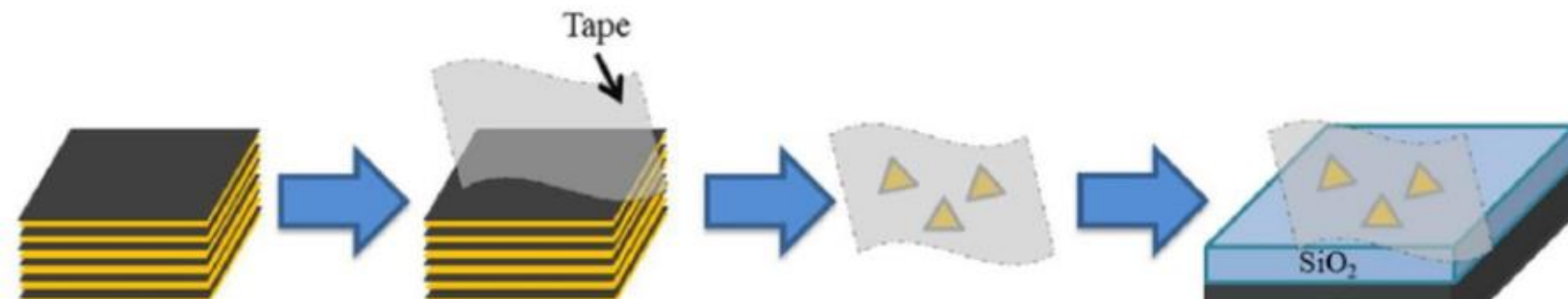
“Studying the superconducting properties of 2D BSCCO yielded inconsistent electrical response due to the **degradation** in properties of few-atom thin BSCCO in an ambient atmosphere. The degradation of superconductivity results in **out-diffusion of oxygen** atoms from the lattice above 200 K. Also, the chemically unstable nature of BSCCO quickly leads to surface **deterioration** by the **adsorption of water molecules.**”

Vilas et al. Scientific Report(2024)



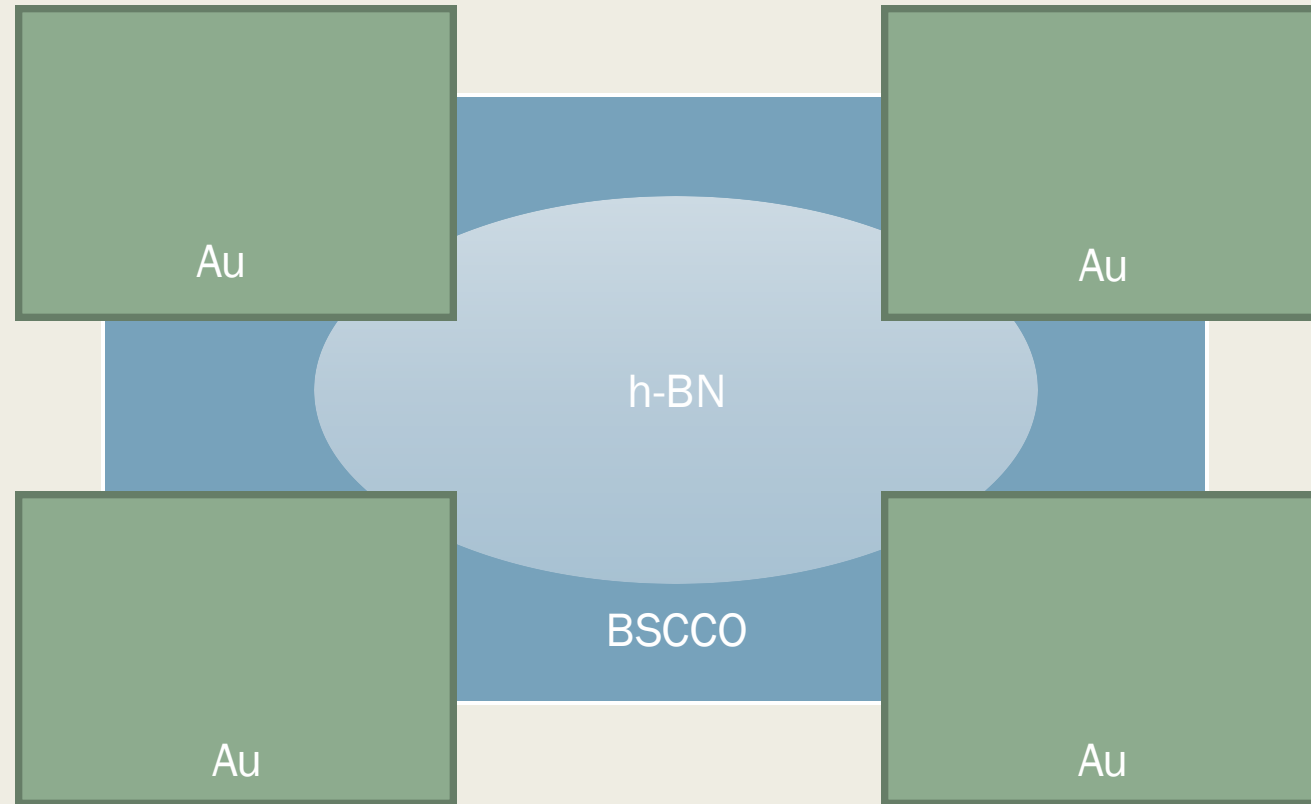
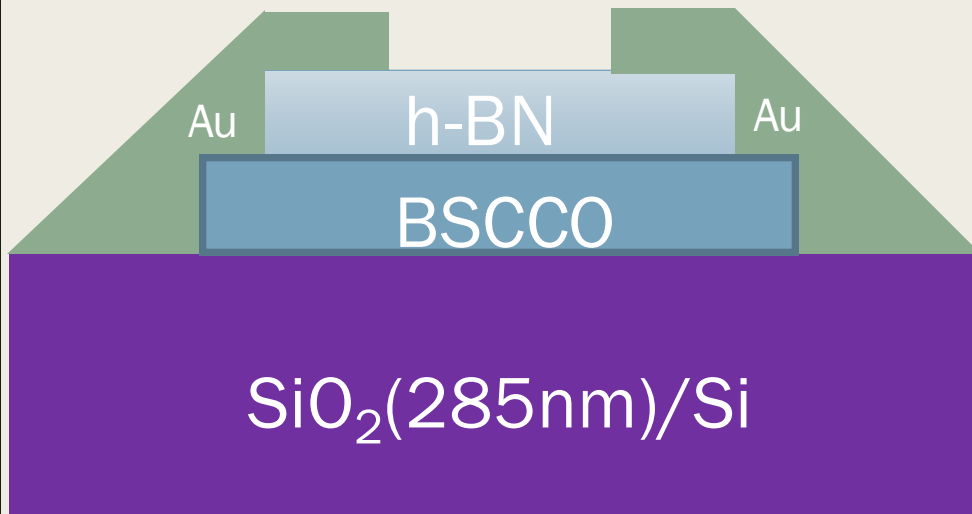
2D Maters Ignacio et al. (2024)

Scotch Tape Method

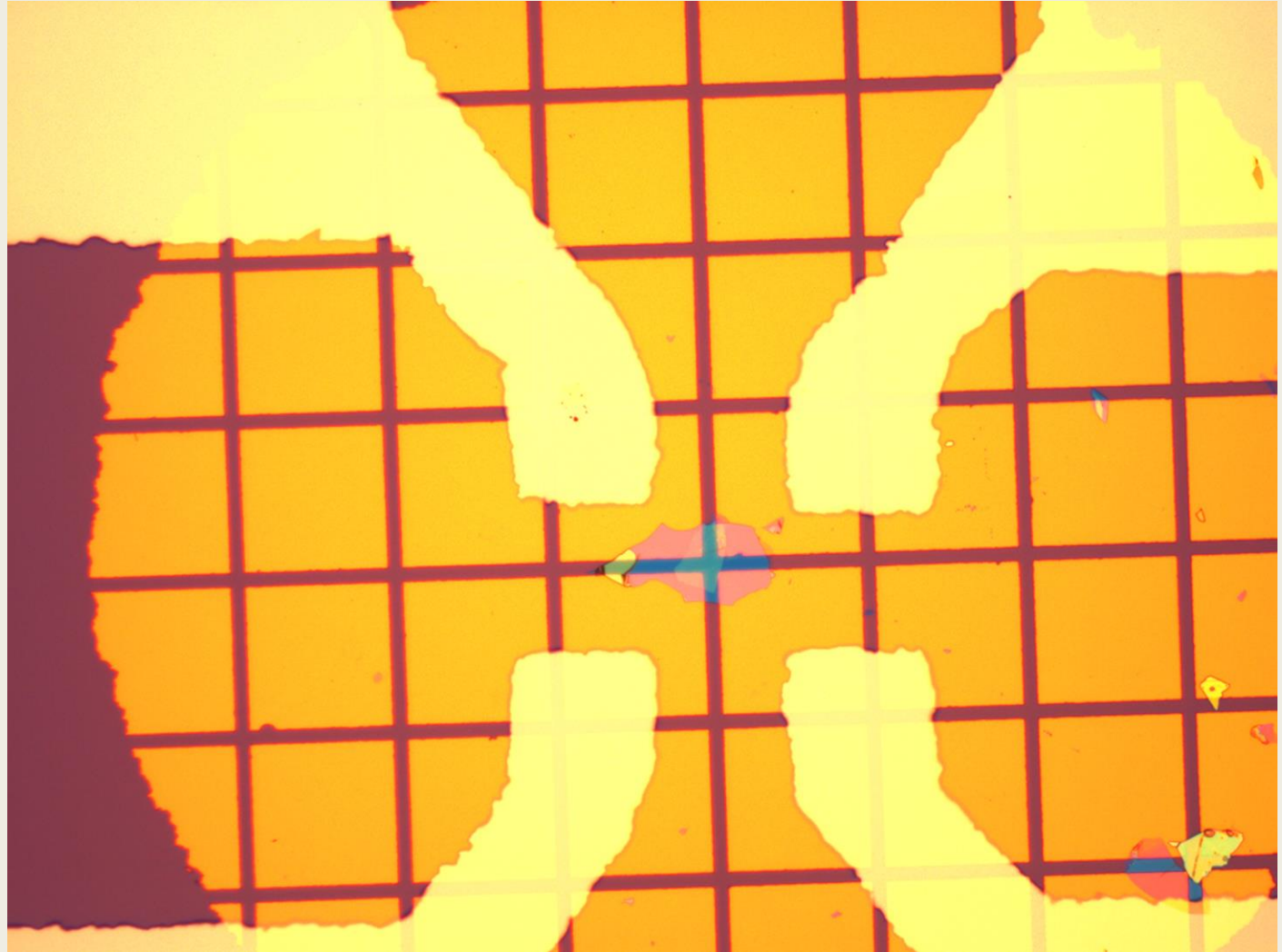


MECHANICAL EXFOLIATION WITH MATERIAL

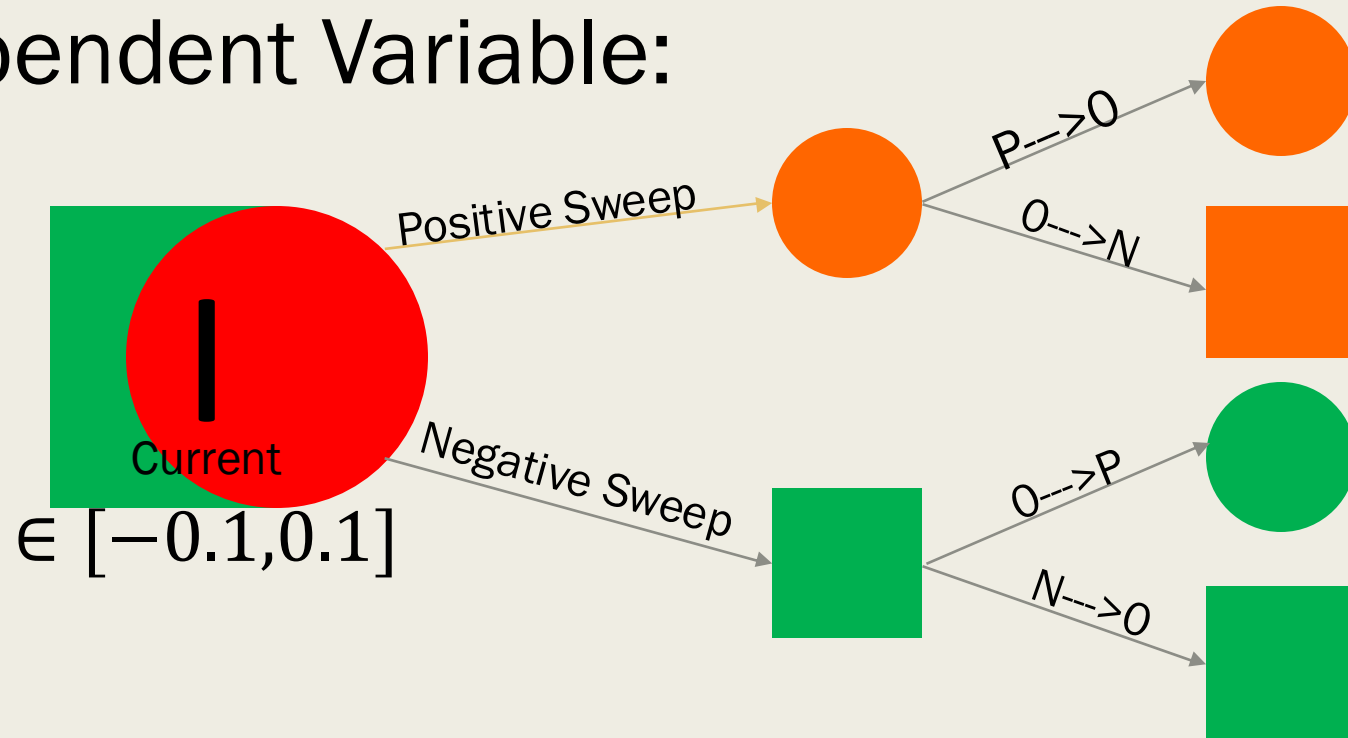
Device Structure



Images



Independent Variable:



Start Current	End Current
+0.1 mA	0
0	-0.1mA
0	+01mA
-0.1mA	0

$$T \in [20K, 110K]$$

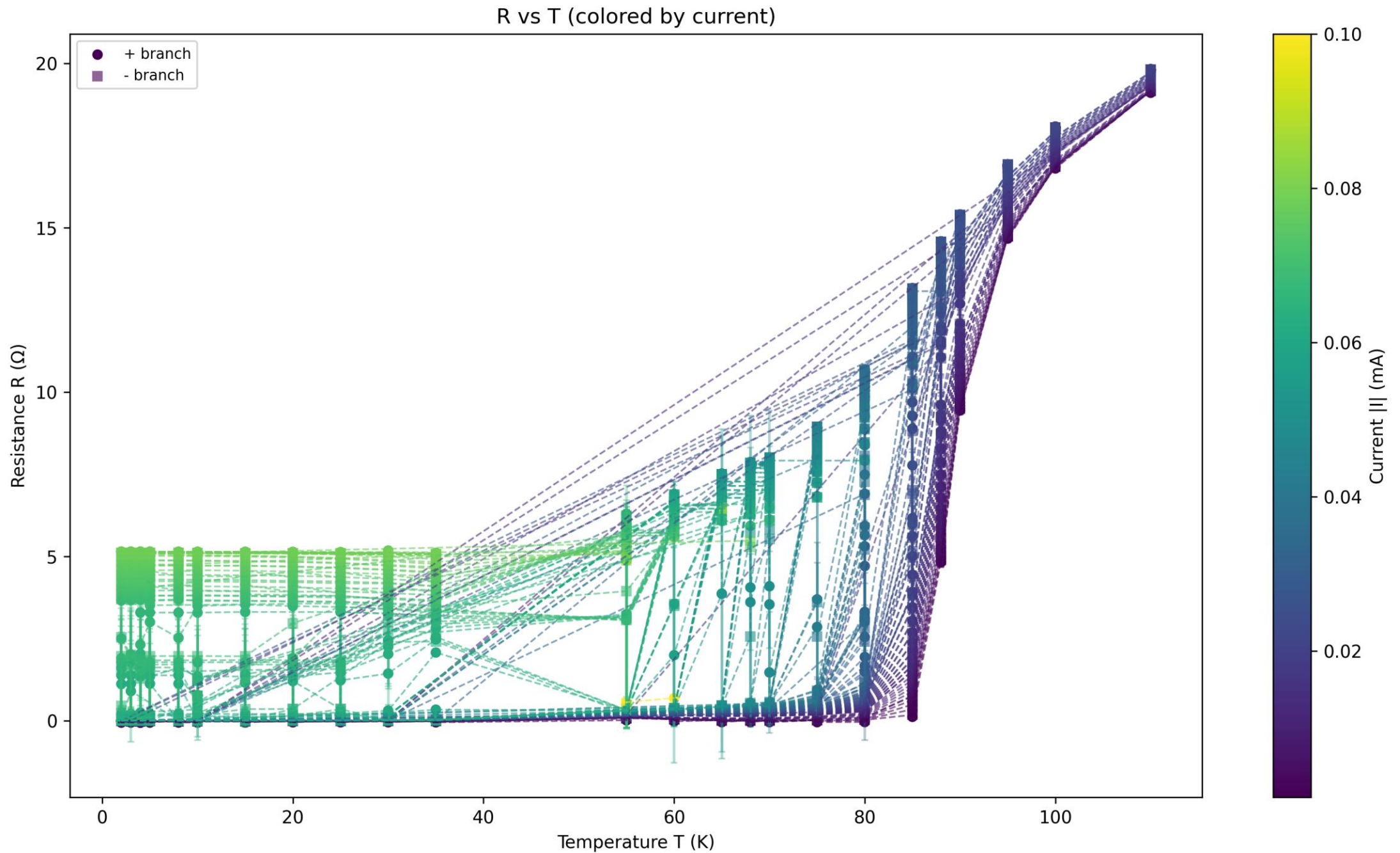
Different Temperature multiple times

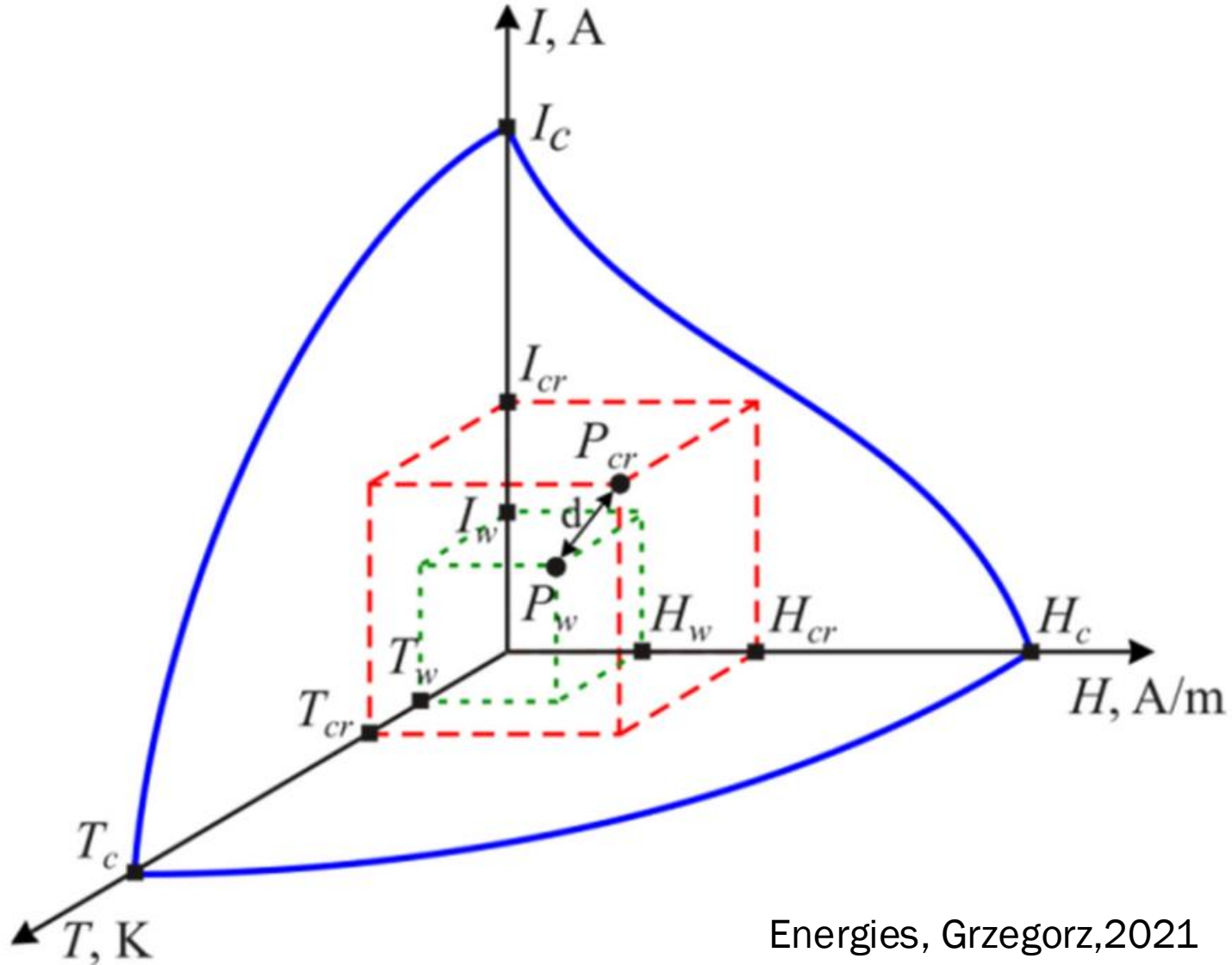
Dependent variable:

$$V \in [-0.4mV, 0.4mV]$$

$$R \in [0, \infty)$$

R-T

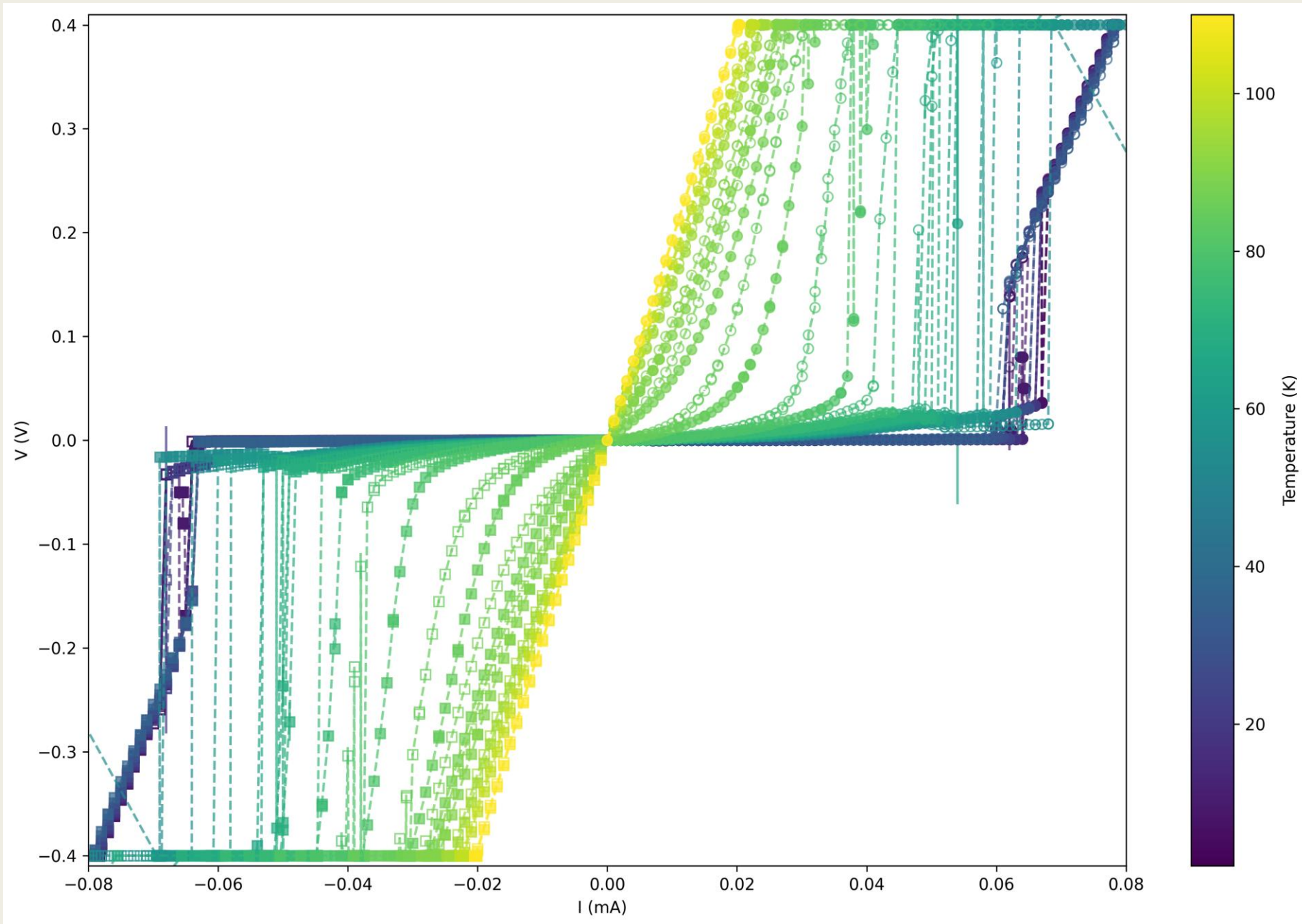




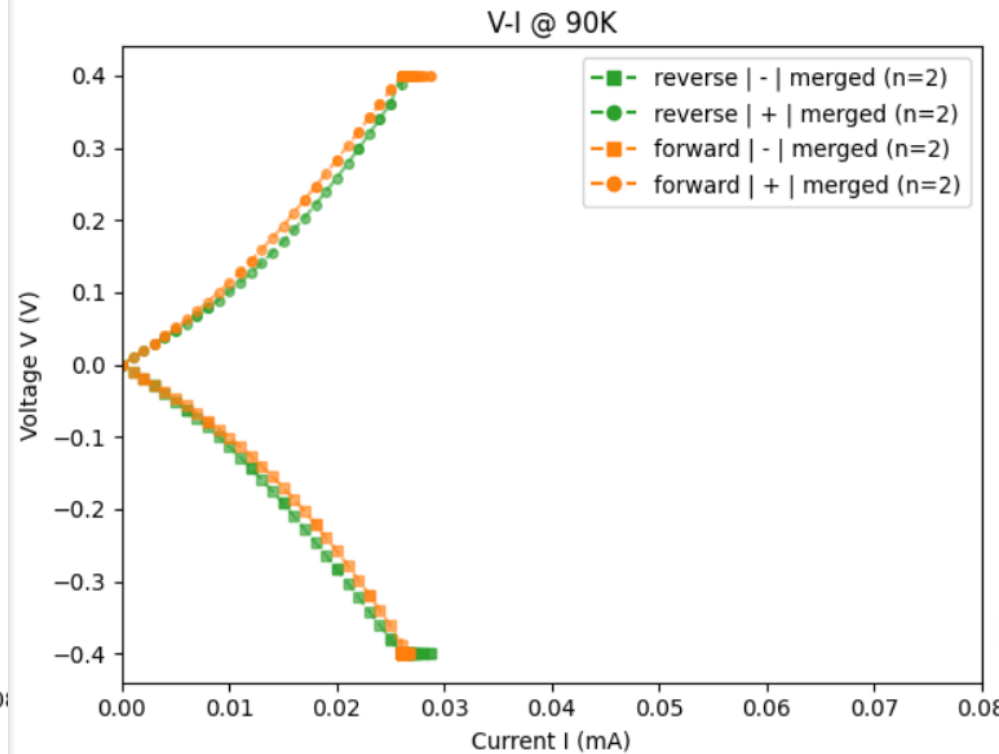
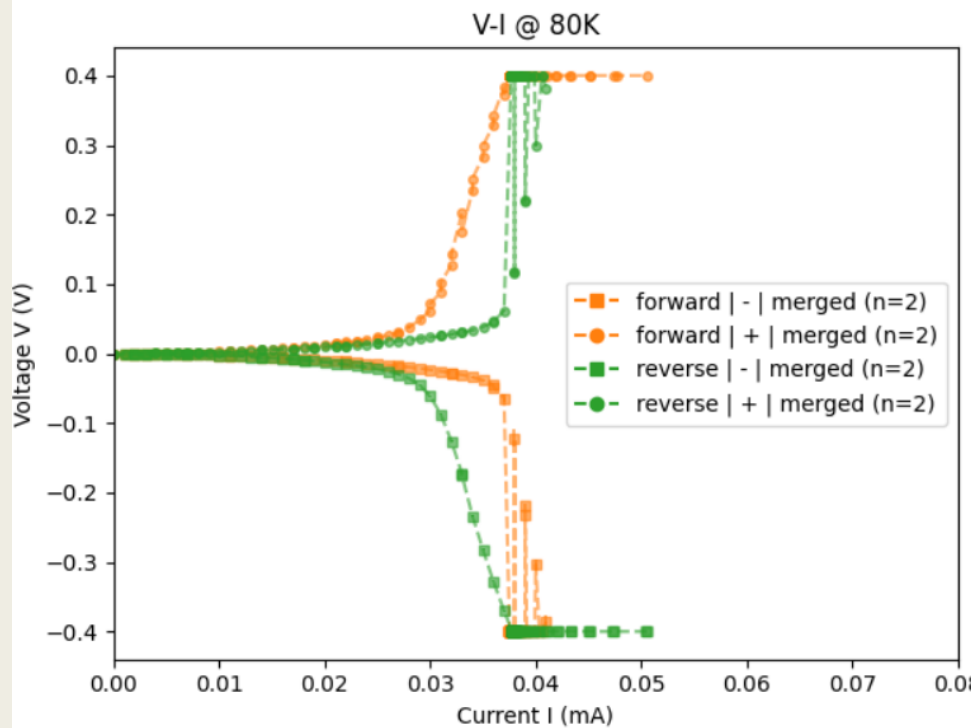
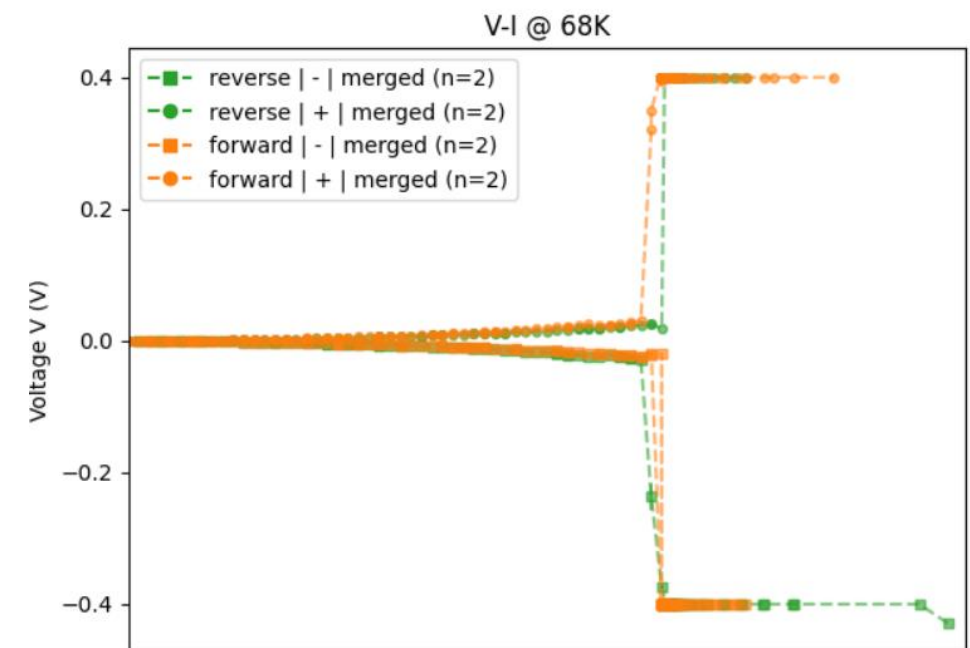
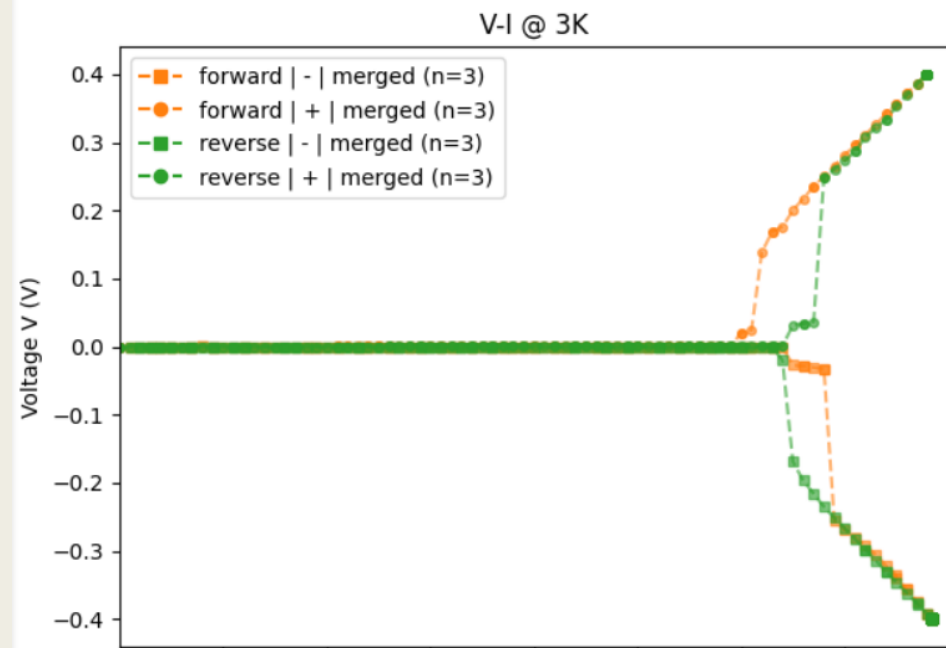
Energies, Grzegorz, 2021

Super Conducting
Critical Surface

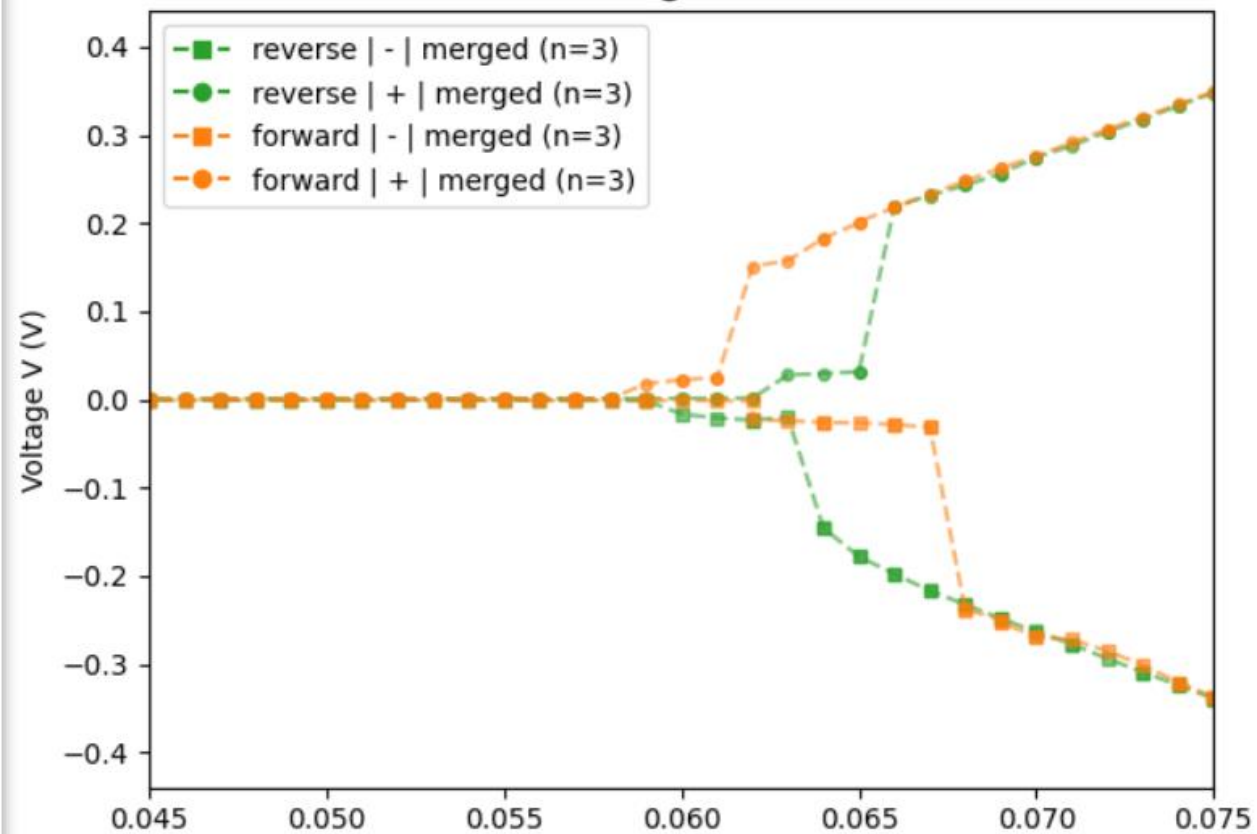
Overview for V-I



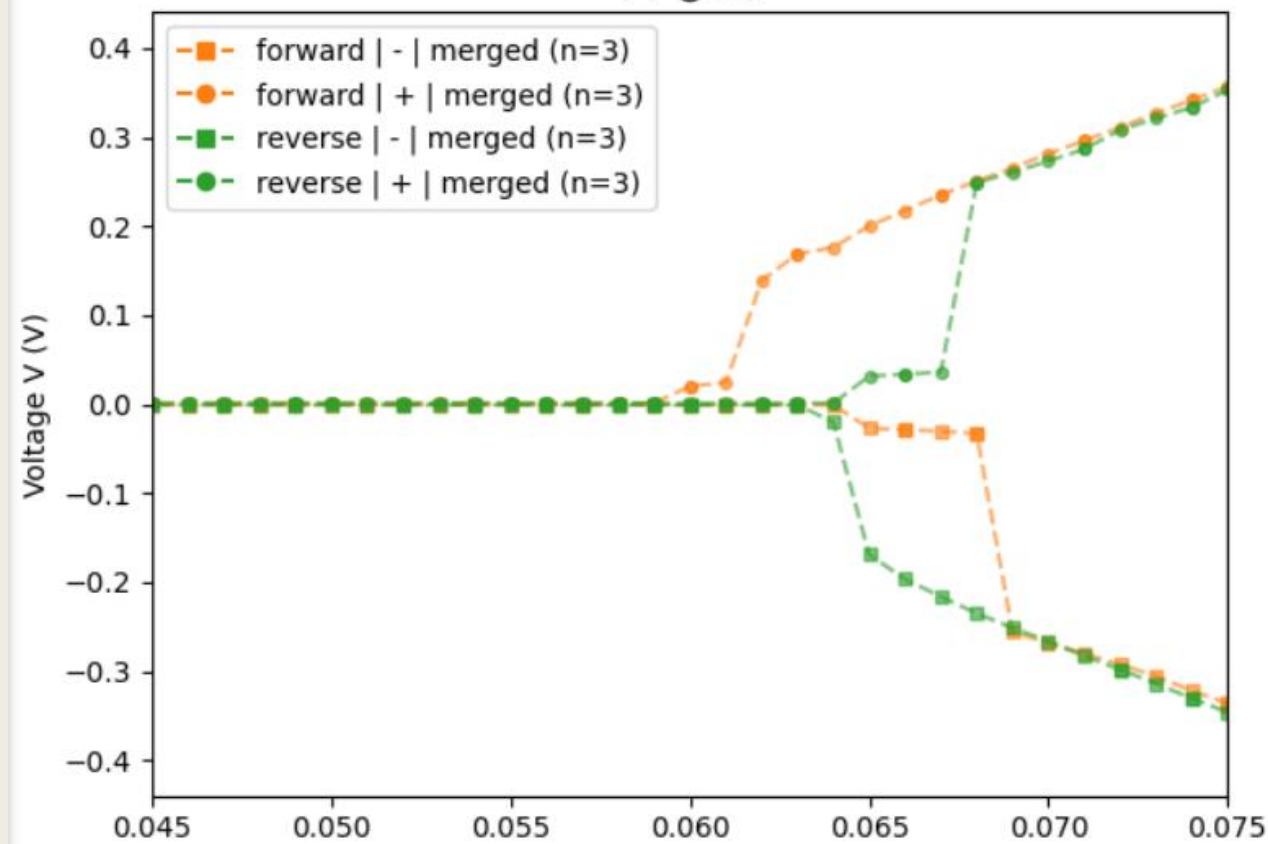
Close look for V-I

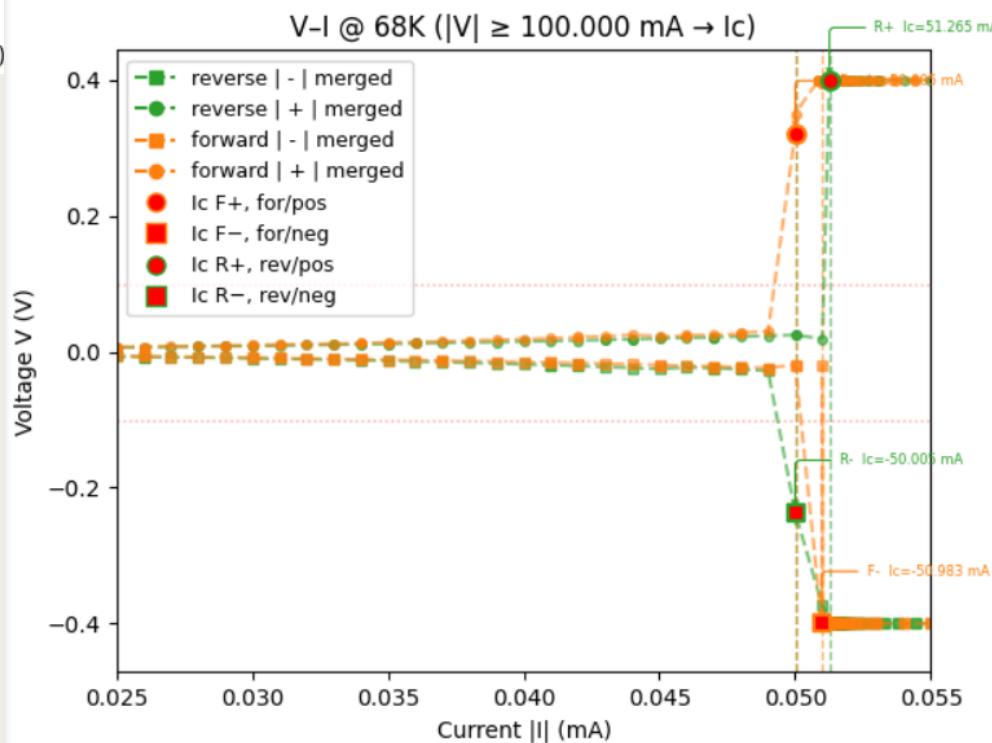
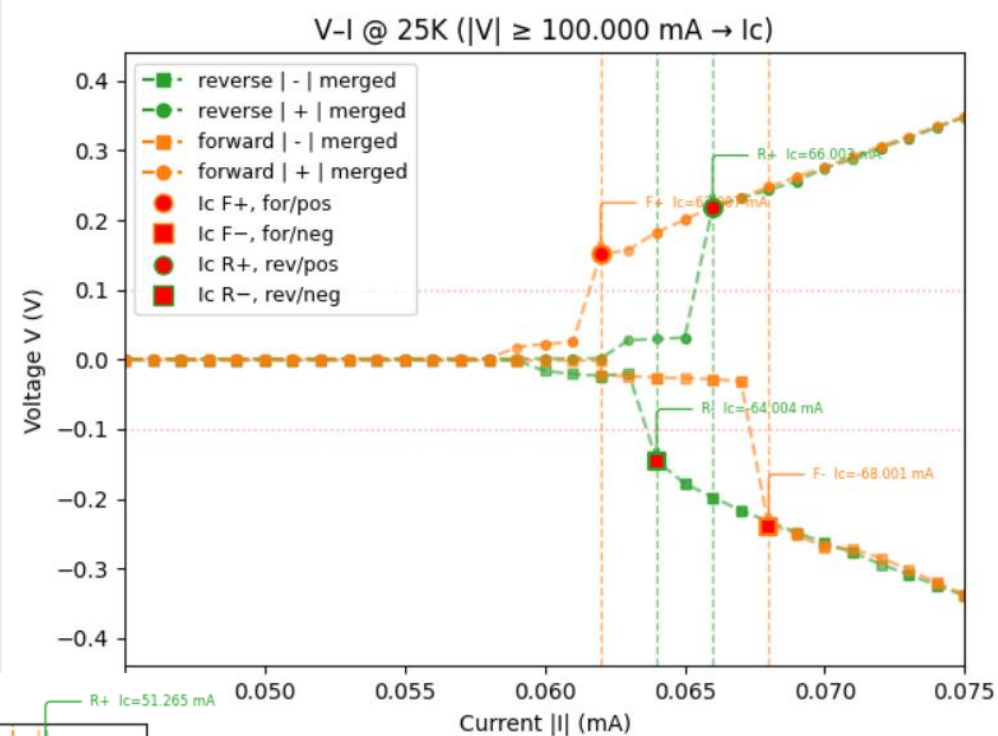
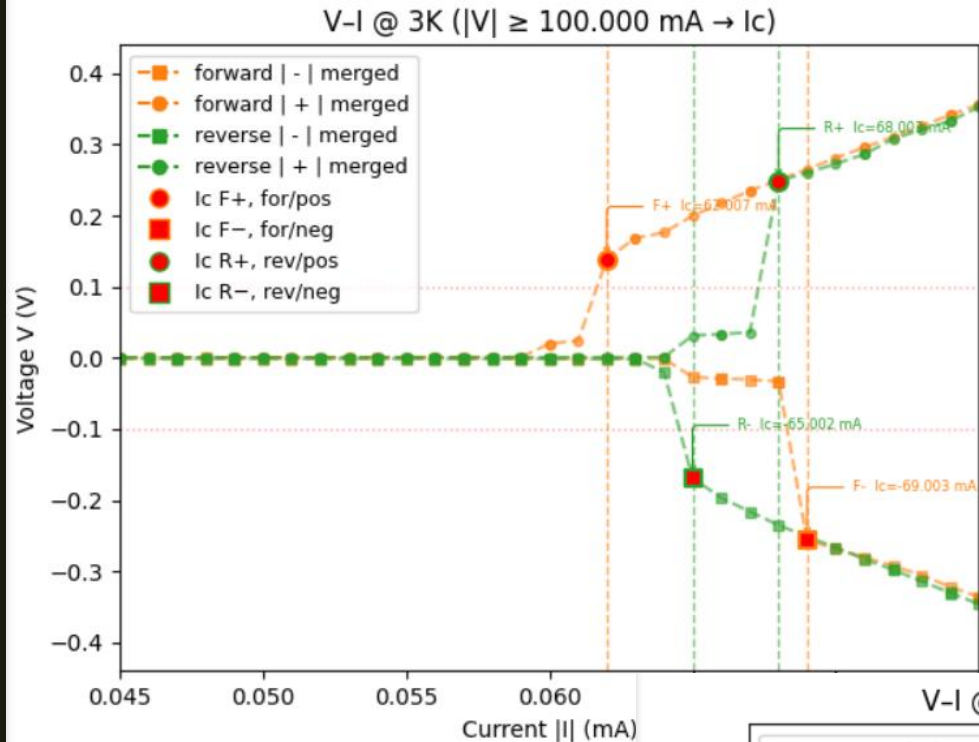


V-I @ 25K

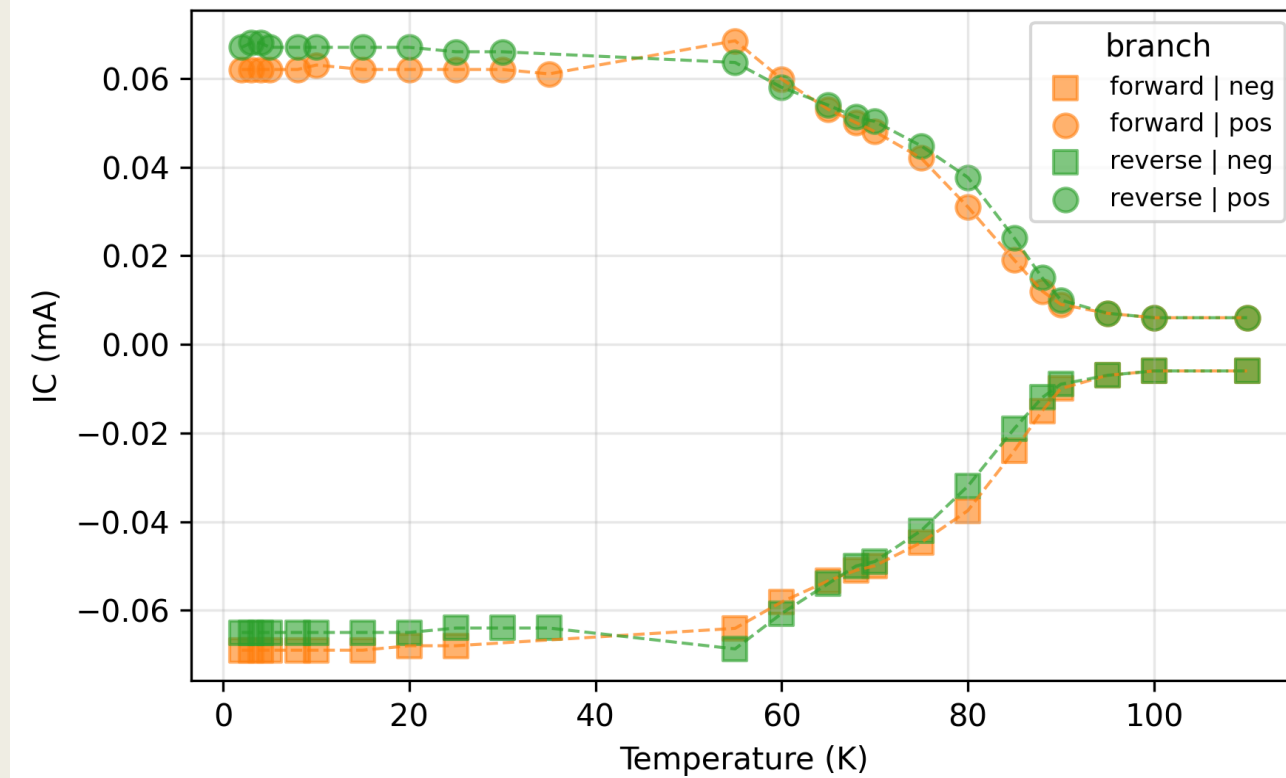


V-I @ 3K

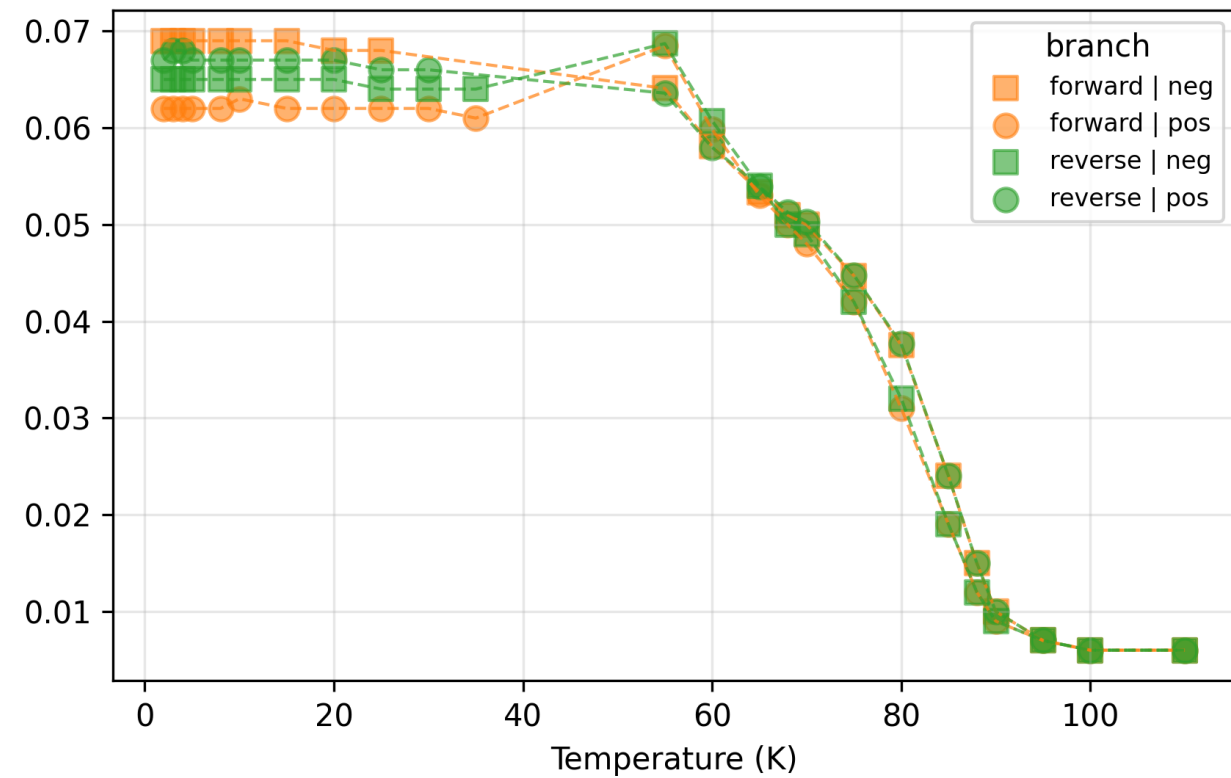




Critical Current vs Temperature



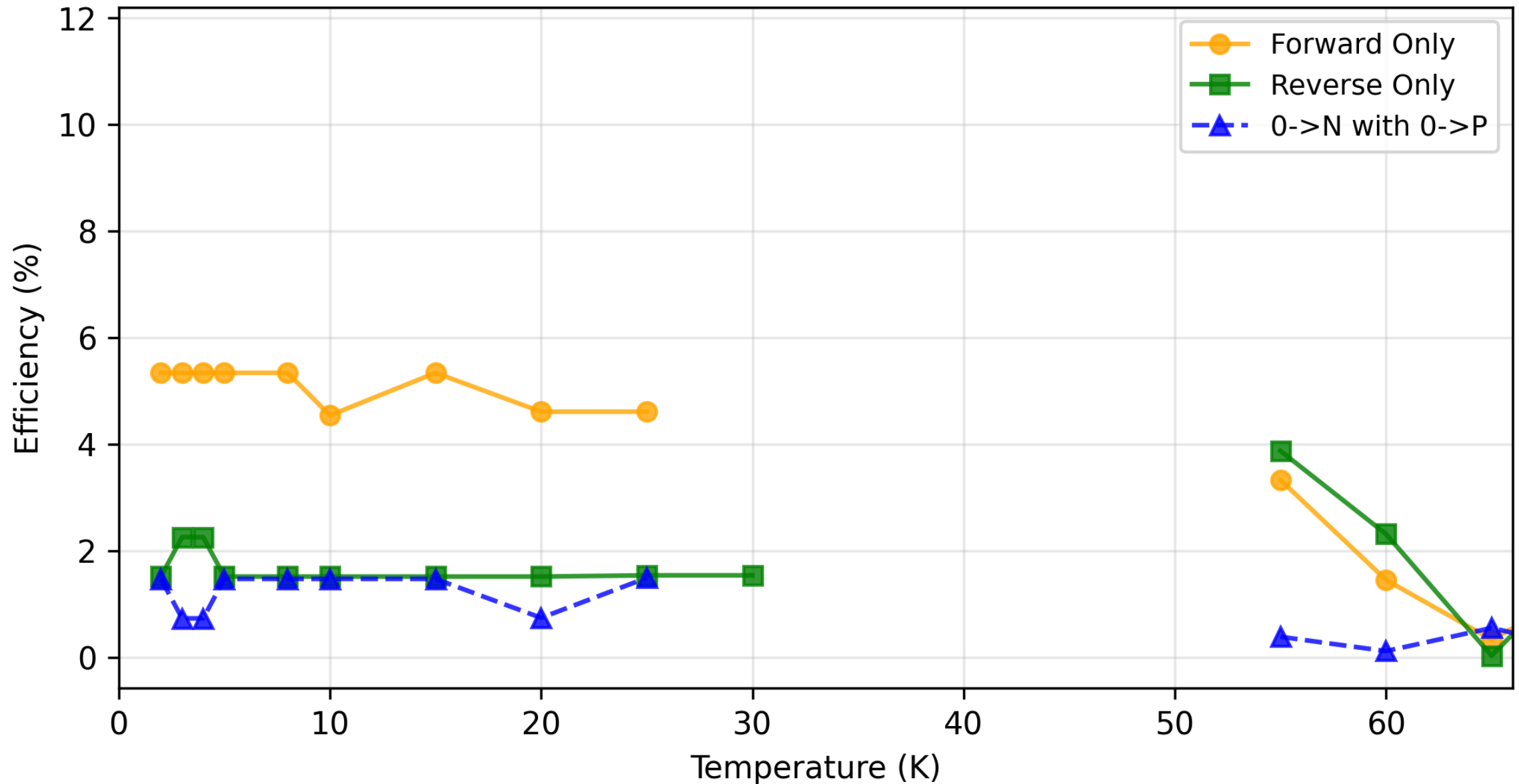
Critical Current vs Temperature



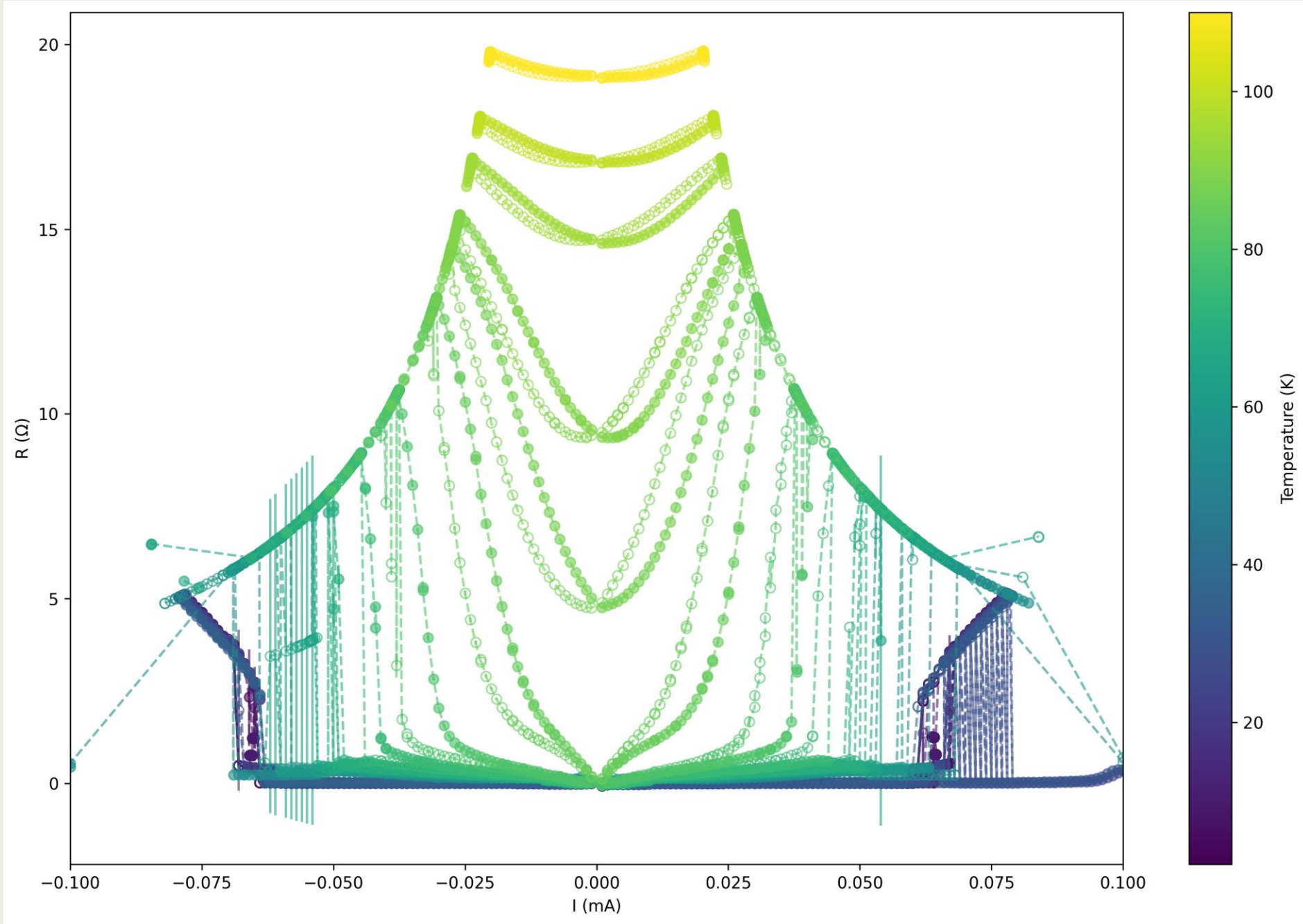
Superconducting diode effect

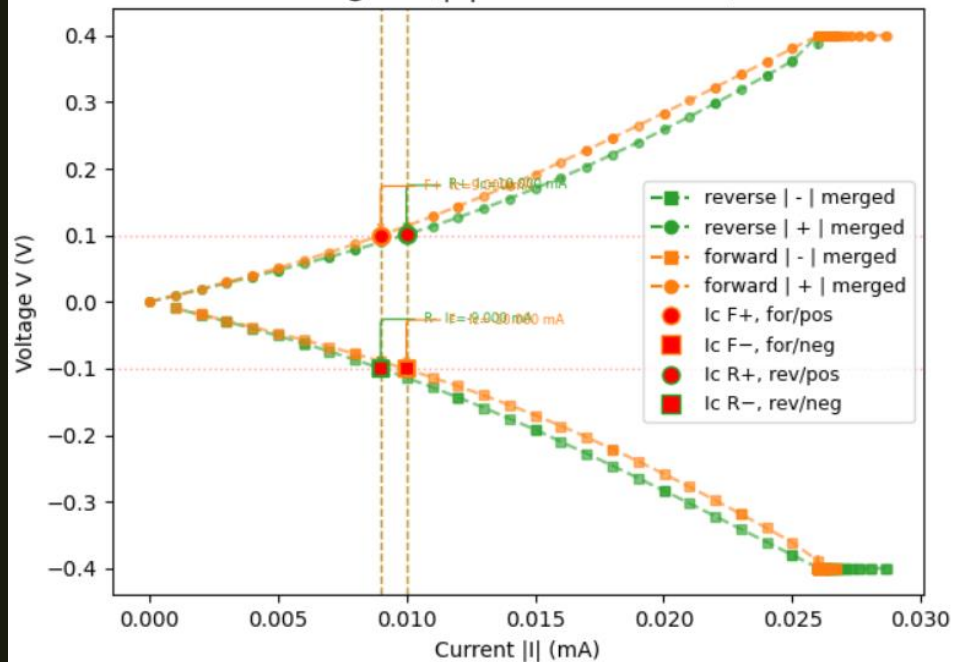
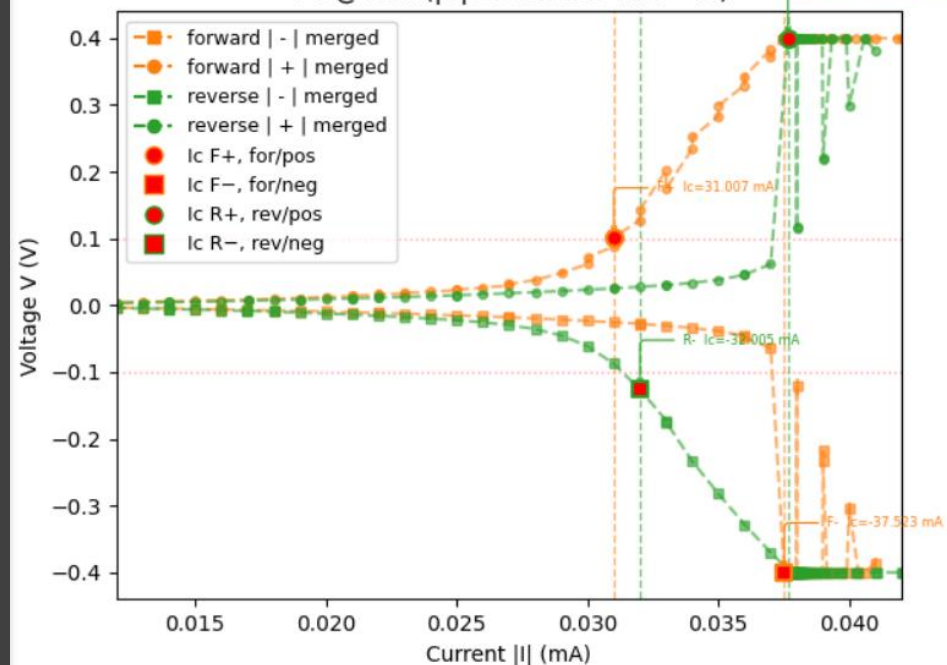
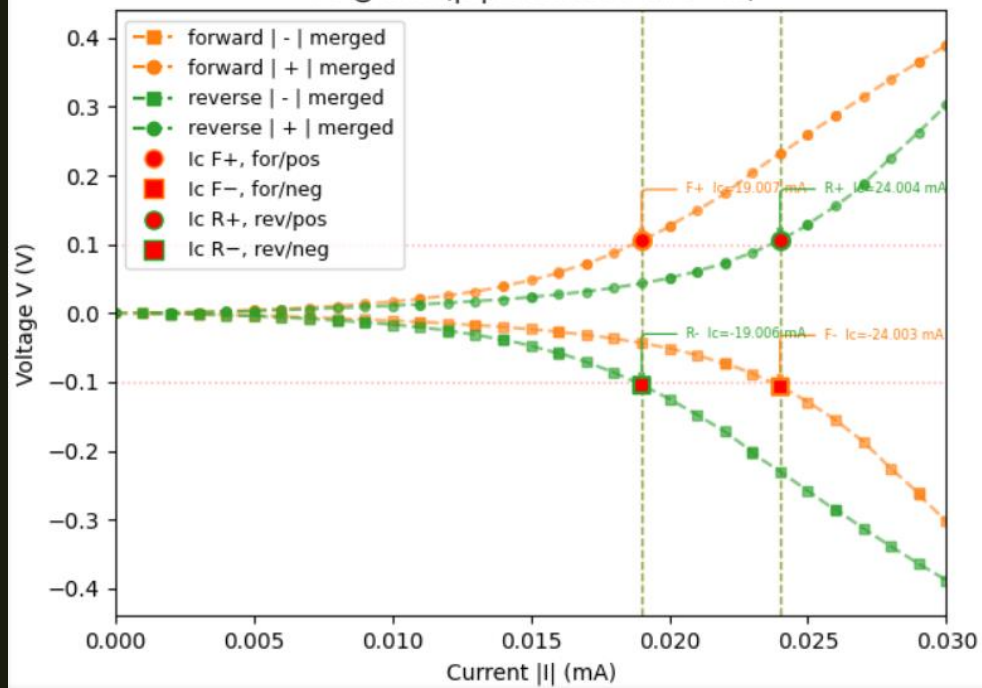
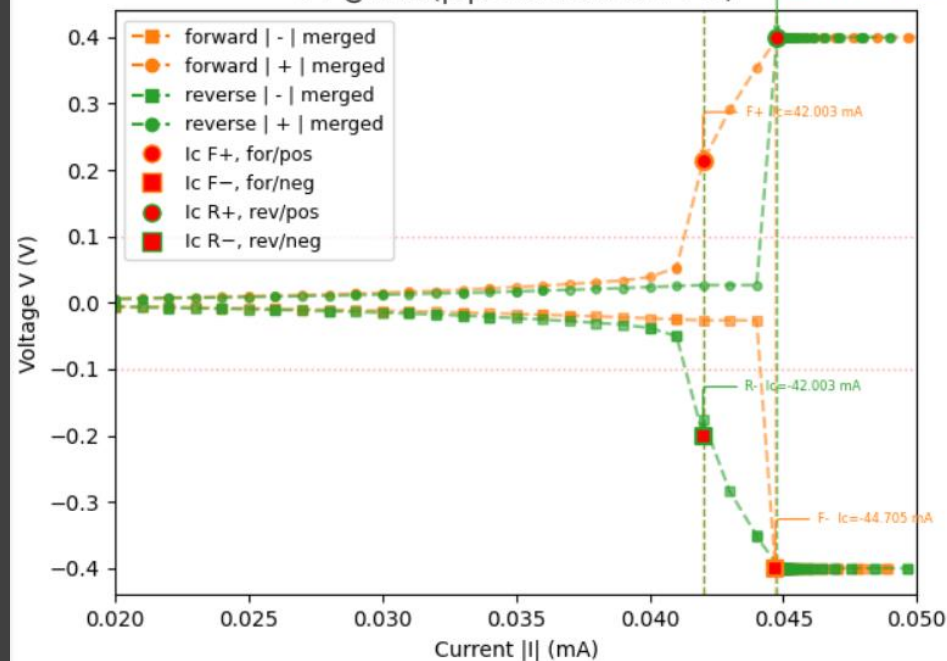
$$\text{Efficiency} = (I_{c+} - |I_{c-}|) / (I_{c+} + |I_{c-}|)$$

Diode Rectification Efficiency vs Temperature

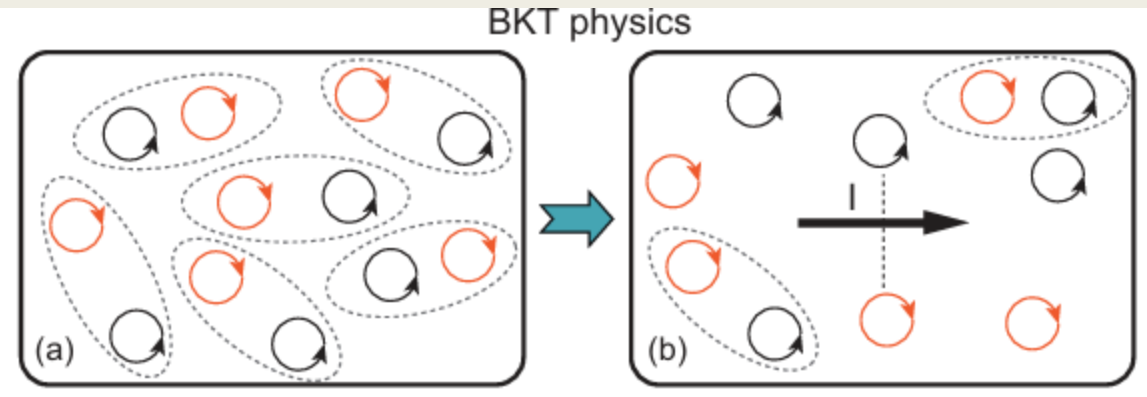
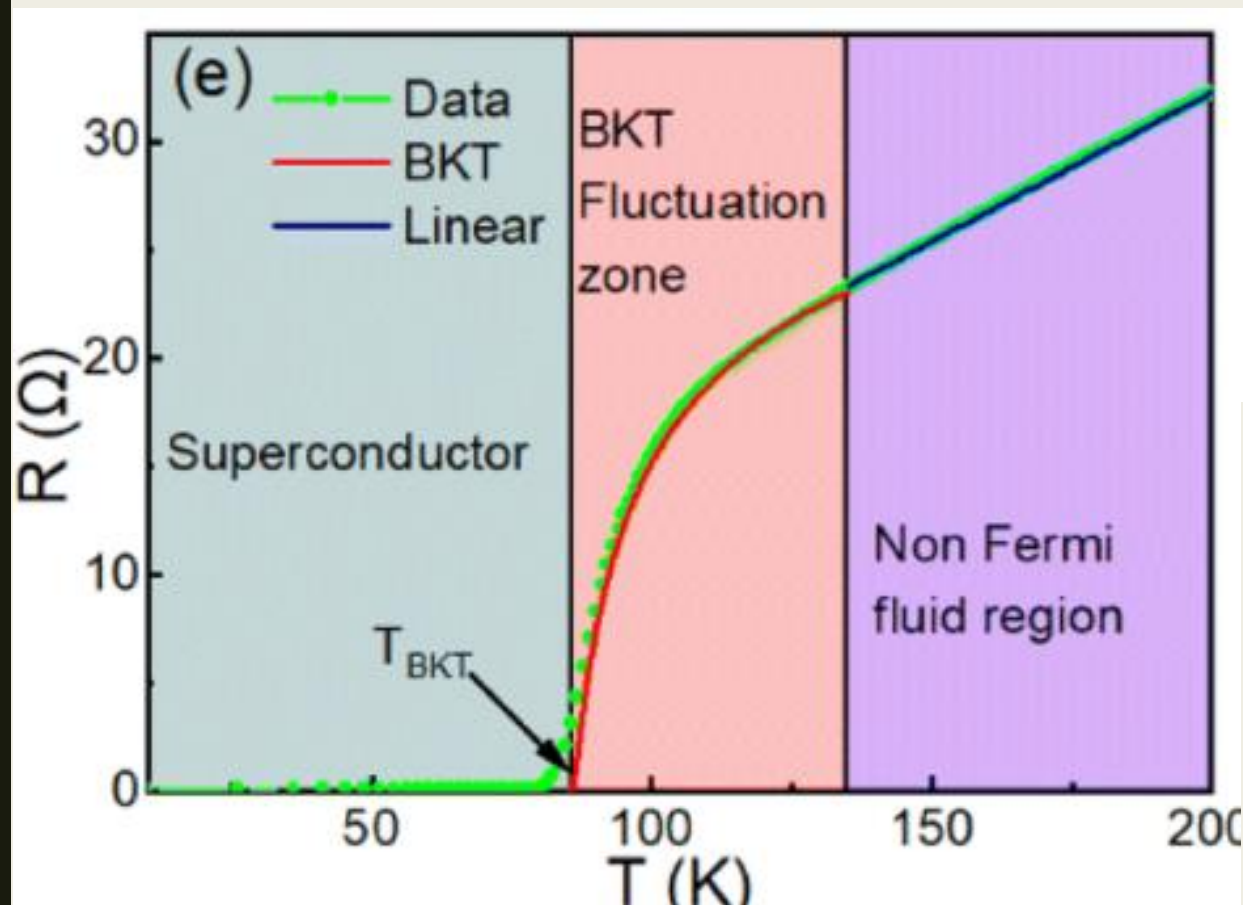


Hysteresis



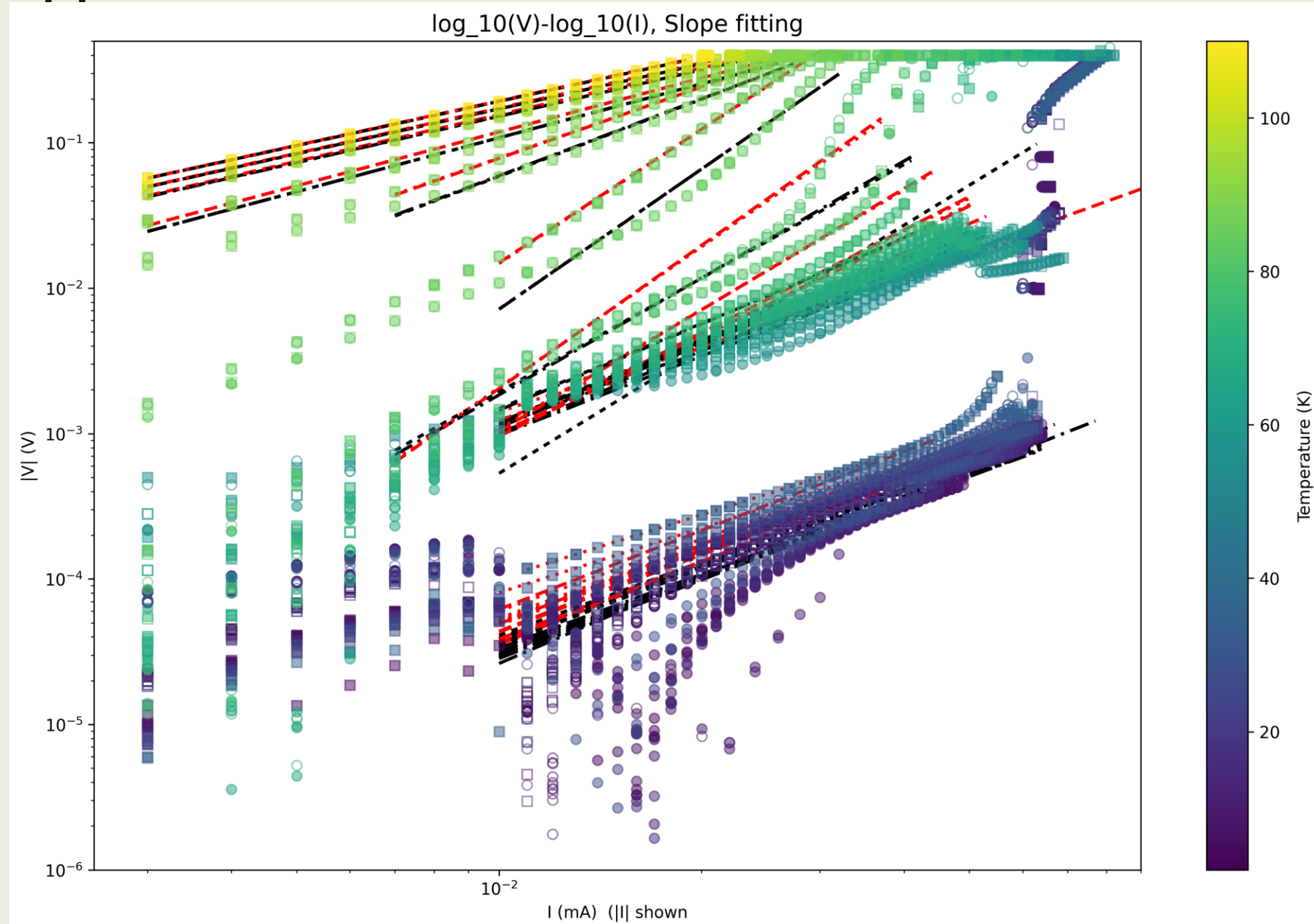
V-I @ 90K ($|V| \geq 100.000 \text{ mV} \rightarrow I_c$)V-I @ 80K ($|V| \geq 100.000 \text{ mV} \rightarrow I_c$)V-I @ 85K ($|V| \geq 100.000 \text{ mV} \rightarrow I_c$)V-I @ 75K ($|V| \geq 100.000 \text{ mV} \rightarrow I_c$)

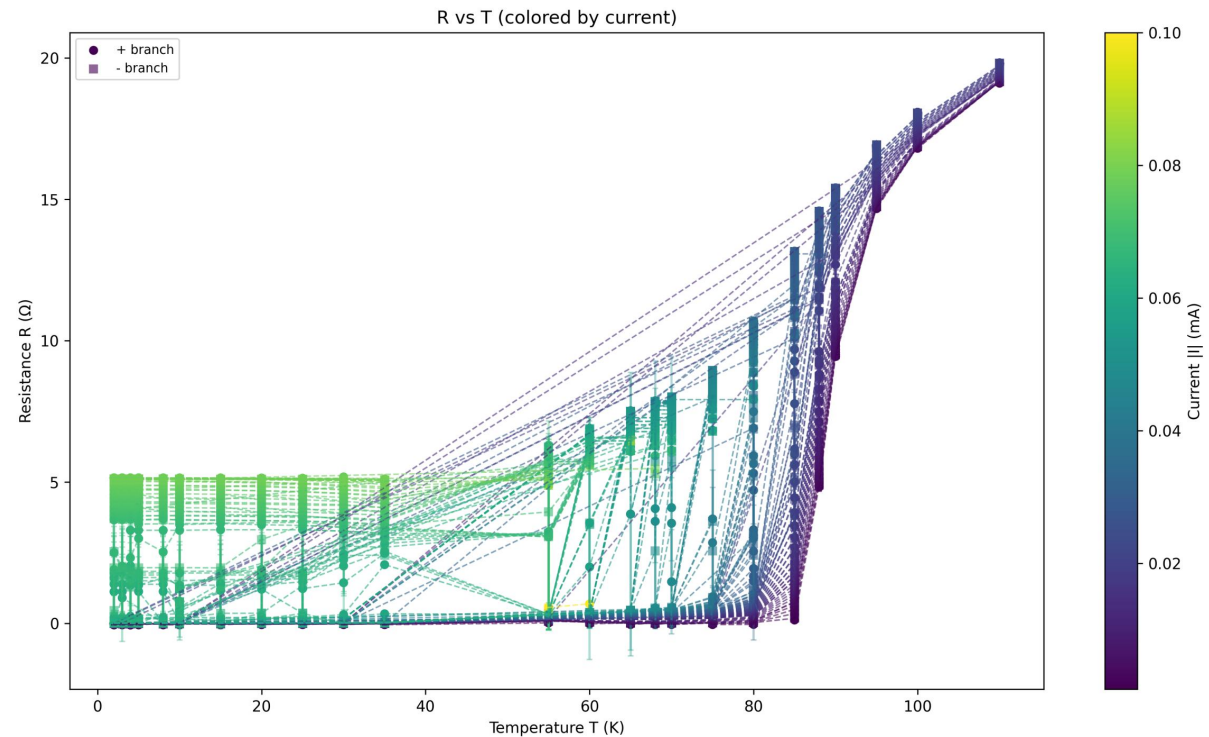
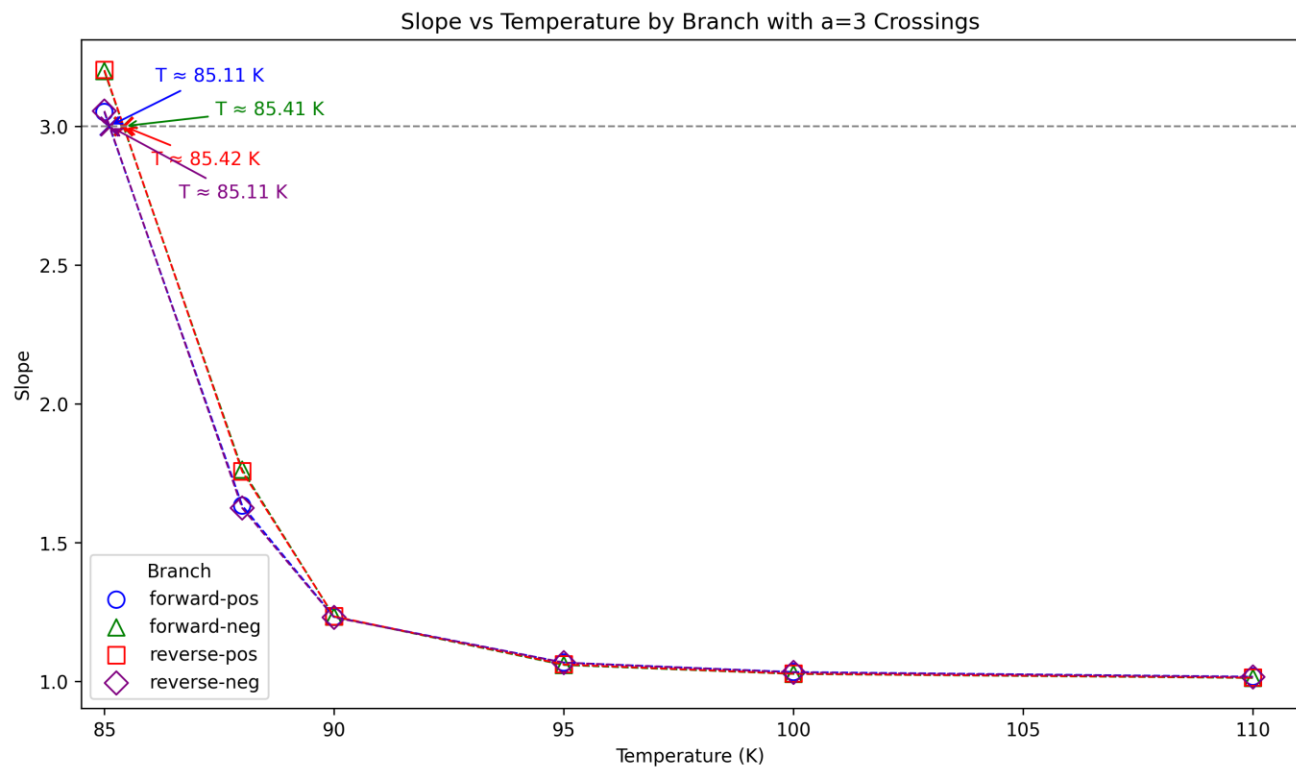
Berezinskii–Kosterlitz–Thouless (BKT) transition

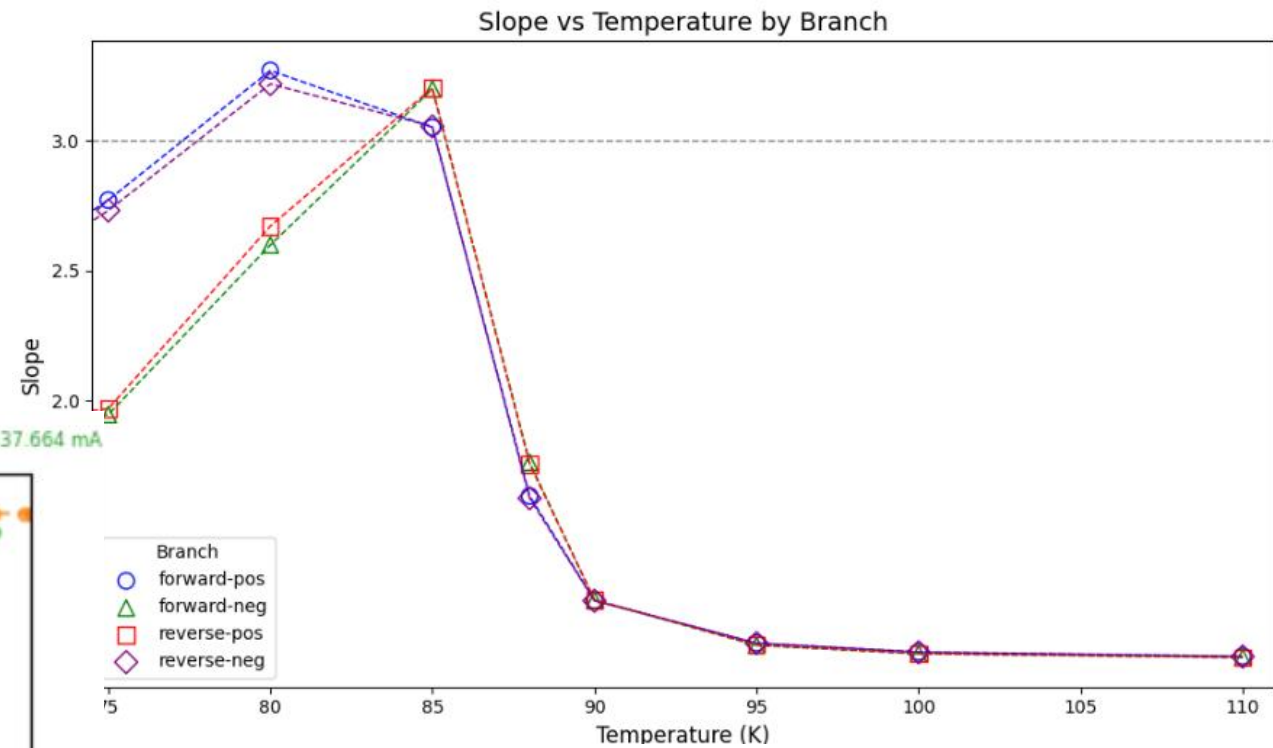
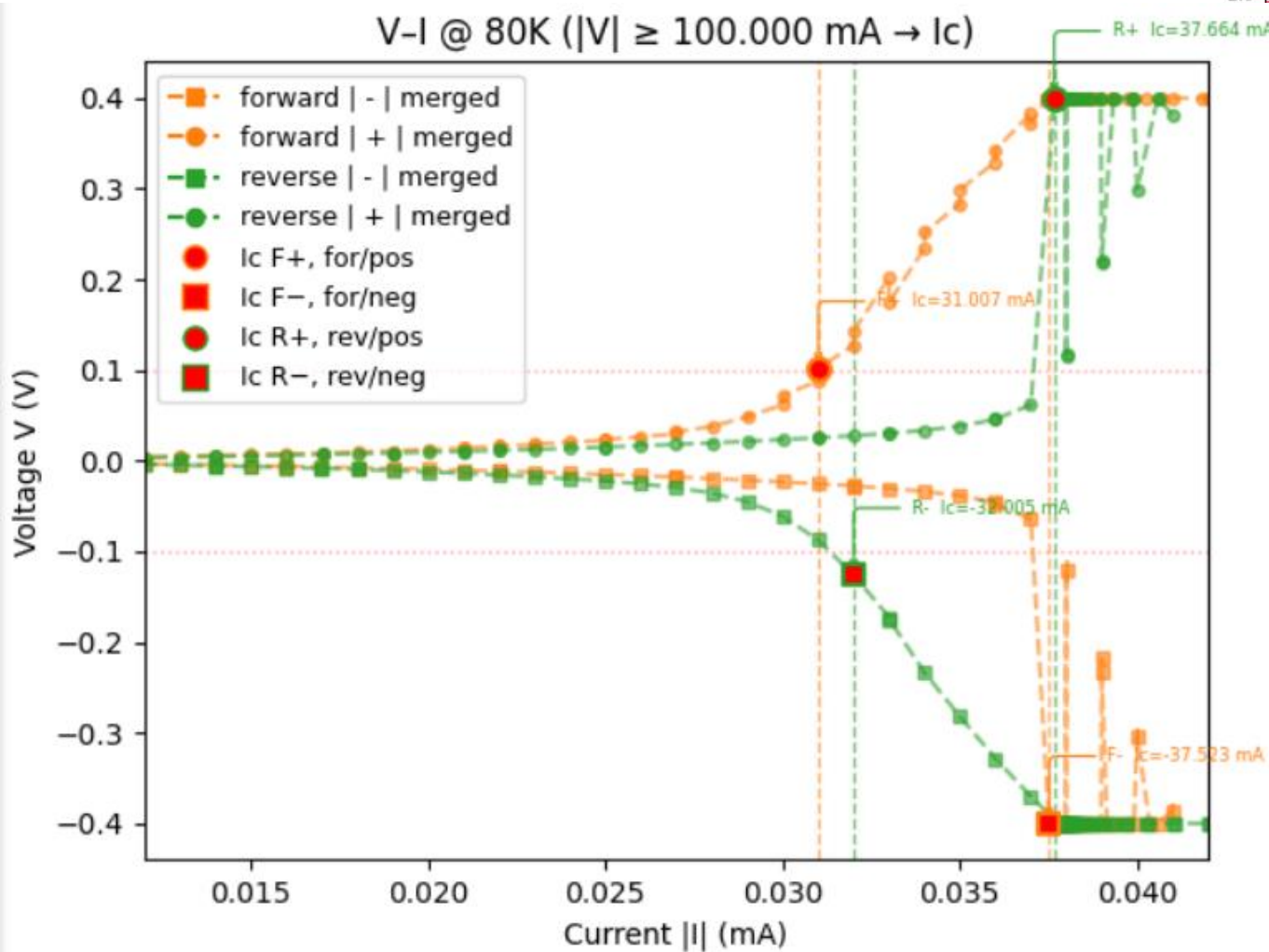


$$V \propto I^{a(T)}, \quad a(T) = \frac{\pi J_s(T)}{T} + 1$$

T_BKT Fitting







Summary

Future Works